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Tai Chi and other mind-body interventions for cancer-related fatigue: an updated systematic review and network meta-analysis protocol.

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Manuscripts

Tai Chi and other mind-body interventions for cancer-related fatigue: an updated systematic review and network meta-analysis protocol.

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ABSTRACT

Introduction Fatigue is one of the most common symptoms in patients with cancer, which is responsible for reduced quality of life. Mind-body interventions, as the nonph armacologic therapy, have the strongest evidence base for treating cancer-related fatigue (CRF). However, the efficacy of Tai Chi in the treatment of CRF is still controversial. Here, we will perform a systematic review and network meta-analysis (NMA) aimed t o assess the relative effects of Tai Chi and other mind-body interventions in patients with CRF.

Methods and analyses We will search the following ten databases from their inceptio n to August 1, 2021: PubMed, EMBASE, Scopus, OVID, Web of Science, Cochrane Central Register of Controlled Trials (CENTRAL), the China National Knowledge Infr astructure (CNKI), China Science and Technology Journal Database (CQVIP), Chinese Biomedical Databases (CBM), and Wan Fang Digital Journals. We will include all ran domized controlled trials (RCTs) comparing Tai Chi with other mind-body intervention s, or with no intervention, placebo, standard care in the treatment of CRF. The primar y outcome will be the changes of fatigue evaluated by effective and validated scales. We will perform a Bayesian NMA to synthesis all evidence for each outcome. Surface

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under the cumulative ranking curve (SUCAR) and the mean ranks will be used to rank the various treatments. Consistency and heterogeneity will be evaluated by employing local and global methods. We will also assess the quality of evidence contributing to network estimates of the outcomes with the Grading of Recommendations Assessment, Development and Evaluation System (GRADE) framework.

Ethics and dissemination The results will be disseminated through peer-reviewed publication. Since no private patient data will be involved in the reporting, there are no ethical considerations associated with this review.

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Strengths and limitations of this study

► This study will include RCTs published in the past 4 years to update the evidence base, and obtain a comprehensive ranking of all included treatments.

►Local and global methods will be employed to evaluate heterogeneity and consistenc y.

►The limitations of primary studies will be evaluated with the revised Cochrane risk of bias tool (RoB 2).

►We will explore whether treatment effects are robust in network meta-regression and assess the quality of evidence using the Grading of Recommendations Assessment, De velopment and Evaluation system (GRADE).

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INTRODUCTION

Cancer-related fatigue (CRF) is defined as 'a distressing, persistent, subjective sense of physical, emotional, and/or cognitive tiredness or exhaustion related to cancer or cance r treatment that is not proportional to recent activity and interferes with usual function ing' by the National Comprehensive Cancer Network (NCCN).[1] It is one of the most prevalent and distressing symptoms of cancer and its treatment which might persist for years after treatment completion in survivors.[2,3] Prevalence estimates of CRF range fro m 14.03% to 100% depending on the latest research evidence.[4,5] CRF has a significant effect on physical functioning during treatment, and it is uncertain whether patients re gain full functioning when treatment is completed. Persistent CRF causes disruption in all aspects of quality of life (QoL) during and after treatment.[6,7] Furthermore, CRF can cause difficulties in end-of-life care and might be a risk factor of reduced survival.[1] Due to few accurate animal models that can mimic human CRF, the exact pathophysi ology of CRF is still unknown. The proposed underlying CRF mechanisms include mi tochondrial dysfunction, peripheral immune activation and inflammation dysfunction, as well as central mechanisms (e.g. neuropeptide, neurotransmitter, and hypothalamic-pituit ary-adrenal (HPA) axis dysfunction).[8,9] To our knowledge, there is no gold standard fo r the management of CRF. Based on the NCCN Guidelines for CRF, some of the spe

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cific nonpharmacologic interventions, such as physical activity, physically based therapies, and psychosocial interventions, have the strongest evidence base for treating CRF.[1] Interestingly, there is increasing interest in mind-body interventions such as Tai Chi in oncology settings, which may provide a new and effective treatment for CRF.[10]

The National Center for Complementary and Integrative Health (NCCIH) defined mind-body interventions as 'techniques designed to enhance the mind's capacity to affect bodily function and symptoms.'[11] Mind-body interventions have shown efficacy in treating common cancer-related side effects, especially in alleviating fatigue symptoms.[12-14] Tai Chi rooted in traditional Chinese medicine and has been practiced for several millennia, which merges physical, spiritual, psychosocial, and behavioral elements to promote human health.[15] Randomized controlled trials (RCTs) indicated that Tai Chi showed significant improvement for fatigue over time in patients with lung cancer and breast cancer.[16,17]

However, the effect of Tai Chi for CRF is still controversial from the evidence of several systematic reviews and meta-analysis.[18-20] The inconsistent results between systematic reviews may be due to differences in the search strategies, inclusion criteria, and comparators in these studies. Furthermore, all studies only reviewed trials before 2016. Recently, new RCTs has been conducted and published.[21-23] Therefore, it is necessary

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to re-evaluate effectiveness of Tai Chi and other mind-body interventions for CRF to present an objective and comprehensive evidence based on the up-to-date resources.

Objectives

This systematic review and NMA aims to assess the relative effects of Tai Chi and o ther mind-body interventions in patients with CRF to better inform clinical practice an d health policies.

METHODS

The study will be conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses for systematic review protocols (PRISMA-P)[24] (http://www.prisma-statement.org/Extensions/Protocols.aspx) and Network Meta-analysis (NMA)[25] chec klist (http://www.prisma-statement.org/Extensions/NetworkMetaAnalysis.aspx). This review is registered on open science framework (https://doi.org/10.17605/OSF.IO/JR2UM), a website for pre-registration of research and disclosure of data and materials.

Criteria for considering studies of this review

Types of studies

We will include RCTs in people with CRF that compared Tai Chi, other mind-body i nterventions against no intervention, placebo, or standard care. No limits will be appli ed for study language, publication status, and publication date. Cross-over, cluster and

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quasi-randomized trials (such as those allocating by using birthday, or the date of hos pital visiting) will be excluded.

Types of participants

Patients with a diagnosis of cancer and symptoms of CRF will be considered. No rest rictions will be set for gender, age, cancer type and grade, during or post-treatment.

Types of interventions

Mind-body therapies were the experimental interventions and Tai Chi is our primary o bservation target. Mind-body interventions in this review include Tai Chi, Qigong, min dfulness-based stress reduction, mindfulness-based cognitive therapy, mindfulness-based interventions, yoga, graded exercise, strengthening exercise, physical training, progressiv e muscle relaxation, resistance training, meditation, and cognitive behavioral therapy (C BT). RCTs in which more than two intervention groups in line with the inclusion crit eria will be kept in this review after the exclusion of groups that did not meet the in clusion criteria.

Types of comparators

Information about the interested interventions will be obtained either from head-to-head or placebo RCTs. Participants in the control group should be treated with no intervent ion, placebo, standard care, or all the interventions listed above.

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Types of Outcome measures

We will extract the effect sizes at the first time point after the end of the interventio ns and the later follow-up time points. Only RCTs that include the following primary outcome can be included.

Primary outcome

Assessments of CRF involved effective and validated scales such as Brief Fatigue Inv entory (BFI), Multidimensional Fatigue Symptom Inventory-Short Form (MFSI-SF), Fu nctional Assessment of Chronic Illness Therapy-Fatigue Survey (FACIT-F), and Functi onal Assessment of Cancer Therapy-Fatigue (FACT-F).

Secondary outcomes

The secondary outcomes will include quality of life and adverse events.

Search methods for identification of studies

Published RCTs will be searched in the following electronic databases: PubMed, EMB ASE, Scopus, OVID, Web of Science, Cochrane Central Register of Controlled Trials (CENTRAL), the China National Knowledge Infrastructure (CNKI), China Science and Technology Journal Database (CQVIP), Chinese Biomedical Databases (CBM), and Wa n Fang Digital Journals from the date of their inception to August 1, 2021, with no r estriction of language. We will also do secondary searches in clinical trials records sit

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es (e.g. the NIH clinical registry ClinicalTrials. gov, the International Clinical Trials Registry Platform (ICTRP), the Australian New Zealand Clinical Trials Registry, and the Chinese clinical registry). The electronic search will be supplemented with manually search the reference lists of included studies for additional studies.

The medical search headings (MeSH) terms and their synonyms (free text) will be combined using the Boolean operators "AND" and "OR". The following MeSH terms will be used: "cancer", "tumor", "carcinoma", "neoplasm", "fatigue", "Tai Chi", and the mind-body interventions mentioned above. The search strategy for EMBASE is shown in **Table 1**.

Table 1
Search Strategy for Embase

#1	'neoplasm'/exp
#2	cancer:ab,ti OR tumor:ab,ti OR neoplasm:ab,ti OR carcinoma:ab,ti
#3	#1 OR #2
#4	'fatigue'/exp
#5	fatigue:ab,ti OR exhaustion:ab,ti OR lassitude:ab,ti OR tiredness:ab,ti OR weariness:ab,ti OR tired:ab,ti
#6	#4 OR #5
#7	#3 OR #6
#8	'tai chi'/exp OR 'exercise'/exp OR 'yoga'/exp OR 'qigong'/exp OR 'mindfulness meditation'/exp OR 'training'/exp OR 'muscle relaxation'/exp OR 'cognitive behavioral therapy'/exp
#9	'tai chi':ab,ti OR taijiquan:ab,ti OR 'tai chi quan':ab,ti OR qigong:ab,ti OR yoga:ab,ti OR exercise:ab,ti OR 'mindfulness-based interventions':ab,ti OR 'mindfulness based':ab,ti OR 'muscle relaxation':ab,ti OR training:ab,ti OR meditation:ab,ti OR 'cognitive behavioral therapy':ab,ti OR 'music therapy':ab,ti OR massage:ab,ti OR tuina:ab,ti
#10	#8 OR #9
#11	#7 AND #10
#12	'randomized controlled trial'/exp OR 'controlled clinical trial'/exp

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#13	'randomized controlled trial':ab,ti OR 'controlled clinical trial':ab,ti
#14	#12 OR #13
#15	#11 AND #14

Data collection and analysis

Selection of studies

Two researchers (HJF and HZ) will independently conduct the selection of studies using the bibliographic software EndNote (https://www.endnote.com/). Initially, duplicate studies will be filtered out. The researchers then read the titles and abstracts to exclude studies that do not satisfy the eligibility criteria. After this selection, the full text of all remaining articles will be obtained to determine which to include. In case of duplicated studies were published, we will include only the trial with the most informative data. A third researcher (QWH) is responsible for reaching consensus with the two authors on the inclusion or exclusion of each study. Finally, we will provide a list of excluded studies and justify the exclusions. The selection procedure will be shown in a PRISMA flow chart (**Figure 1**)

Data extraction

Two researchers (YT and DZ) will independently extract data using ADDIS (http://www.drugis.org/index) and Excel software with respect to five main domains: study information (e.g. title, source of publication, year of publication, first author's name and af

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filiation, country), participant information (e.g. gender, age, setting, cancer type, tumor grade, and basic cancer treatment plan), intervention details (e.g. intervention type, duration and frequency), methodology information (study design, random sequence generation, allocation concealment, blinding, and other concerns about bias), and outcome measures. The two researchers will cross-check and ascertain that the data are recorded correctly. Disagreements will be resolved by a third researcher (SYD).

Unit of analysis issues

Due to the lack of individual patient data, the unit of analysis will be performed based on aggregated outcome data.

Dealing with missing data

When means, SDs, and numbers of patients in each arm are not reported, we will transform the recorded SEs, t statistics or *p* values to SDs, according to the Cochrane Handbook (https://training.cochrane.org/handbook/current/chapter-06).[26,27] If the information mentioned above is unavailable, we will send a request for missing data to the corresponding authors or relevant authors of the trial. If there is no reply from the authors, RCTs with missing outcome data will be excluded from the meta-analysis.

Quality assessment

Two independent researchers (YT and JL) will assess the risk of bias of the included

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RCTs according to the revised Cochrane risk of bias tool (RoB 2).[28-30] We will rate e ach domain as either low risk, some concerns, or high risk, based on the five distinct domains: the randomization process, deviations from the intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. We will also make an overall risk of bias judgment according the criteria[31]: (1) The study will be judged to be at low risk of bias for all domains for this result; (2) to raise s ome concerns in at least one domain for this result but not to be at high risk of bias for any domain; (3) to be at high risk of bias in at least one domain for this result, or the study is judged to have some concerns for multiple domains in a way that sub stantially lowers confidence in the result. If the researchers disagree, a third researcher (CW) will adjudicate the discrepancies.

Data Synthesis and Analysis

The evidence-base and information flow in the network

We will systematically and comprehensively describe the characteristics of all eligible trials. A network diagram will be used to presented the available evidence.[32] The size of the nodes will represent the sample size of included trials. The thickness of the lin e will reflect the number of studies for each direct comparison, and the color of each edge will imply the risk of bias. For NMAs including many competing interventions a

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nd multi-arm studies, we will instead use a table to present the network structure.[33,34] To understand how much each of the direct comparisons contributes to the final summary data, the contribution matrix will be used to present the percentage information of the direct evidence contributes to each relative effect estimated.[35]

Pairwise meta-analyses

For pair-wise comparison, we will use the DerSimonian-Laird random effects model. Weighted mean difference (WMD) for continuous outcomes or odds ratios (ORs) for dichotomous outcomes, both with 95% confidence interval (CI) will be calculated as effect measure. The Cochran's Q χ^2 test and I^2-statistic will be measured for heterogeneity, as a measure to reflect the underlying differences between the RCTs that directly compare the same pair of interventions. A p value of up to 0.10 and I^2 value of above 50% indicate high heterogeneity.[29] We will perform sensitivity analysis of pairwise meta-analysis to validate the robustness of the results by omitting studies with unacceptable sources of heterogeneity.

Examination of assumptions in network meta-analysis (transitivity, inconsistency, and heterogeneity)

Studies that compare different interventions may differ in a broad range of characteristics, which are sometimes associated with the effect of an intervention and referred to

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as effect modifiers. We will evaluate the transitivity assumption underlying NMAs by comparing the distribution of clinical and methodological variables which can act as e ffect modifiers across treatment comparisons.[26,27,36] Consistency in NMA means that the different sources of evidence (direct and indirect) are consistent with each other. To c heck the assumption of inconsistency, we will employ local as well as global method use and consider their results jointly. A recommended local approach named 'node-spli tting' will be used to investigate inconsistency.[37] We will evaluate and detect global i nconsistency and heterogeneity via inconsistency models with I^2 -statistic and predictive interval plot.[36] If there is inconsistency and considerable heterogeneity in the RCTs, ne twork meta-analysis will not be performed and a narrative systematic review will be p rovided.

Network meta-analyses

All analyses will be performed using the *gemtc*, *netmeta*, and *ggplot* package of R ve rsion 3.5.0, and the *network* package in Stata version 15.1. If the included trials meet the above three assumptions, we will perform a random-effects NMA within a Bayesia n framework. We will fit our model using WinBUGS (version 1.4, three Markov chai ns, 50 000 iterations, an initial burn-in of 10 000, and a thinning of 10) and use uninf ormative prior distributions for the treatment effects. The binomial likelihood will be u

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sed for dichotomous outcomes and the normal likelihood for continuous outcomes.[26,27] ORs or WMD for all pairwise comparisons with 95%CI will be summarized in a leag ue table. We will use the surface under the cumulative ranking curve (SUCRA) and t he mean ranks to rank the various interventions for all outcomes.[38] A SUCRA equal to 1 means that the treatment is considered to be the best, while 0 means that it is c ertain to be the worst.

Publication bias, subgroup analyses, and meta-regression

We will perform contribution plot to evaluate the contribution of each direct comparis on to the assessment of each network meta-analytic summary effect. Additionally, we will use comparison-adjusted funnel plots to detect the potential publication bias in the results between imprecise and more precise trials.[39] To assess whether the treatment ef fects for the primary outcome is impacted by effect modifiers, sub-group analyses and network meta-regression will be conducted according the following characteristics: (1) cancer type; (2) patient status (during or after cancer treatment, or no treatment); (3) r andomization and blinding; (4) sample size (less than 25 patients per intervention arm). The sensitivity analysis of NMA will be narrowed into head-to-head studies or trials with low risk of bias.

GRADE quality assessment

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Two experienced researchers (HZ and DZ) will independently assess the certainty of e vidence contributing to each network estimate of the primary outcome using the gradi ng of recommendations assessment, development, and evaluation (GRADE) framework. We will rates the quality of evidence as high, moderate, low, or very low on the basi s of the study limitations, imprecision, inconsistency, indirectness and publication bias.[40] Discrepancies will be resolved by a third researcher (CW).

Discussion

Although the effect of Tai Chi for CRF is still controversial, Tai Chi has the potentia l to be an effective complementary treatment option for cancer-related fatigue. In this review, we will perform the most comprehensive and up-to-date literature search, inclu ding 10 electronic databases, clinical research registration websites and manual searche s. We will also extract the effect sizes at the end of intervention period and the post-intervention follow-up to evaluate the short- and long-term efficacy. Pairwise and netw ork meta-analyses will be performed. We are confident that this research can update t he evidence of Tai Chi and mind-body interventions in the treatment of CRF, and pro vide evidence-based medicine for patients, clinicians, and policy makers.

Ethics approval statement Since no private patient data will be involved in the repor ting, there are no ethical considerations associated with this review.

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Authors' contributions

HJF and HZ contributed equally. HJF, YT and CW devised the study. HJF drafted thi s protocol. QWH and CW revised the protocol. HJF, HZ and QWH will perform the study search and study selection. YT, DZ, and SYD will carry out the data collection. YT, JL, and CW will assess the quality of included RCTs. HJF and YT will conduct the data analysis. All authors will provide input on the final draft of the protocol.

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Competing interest statement None declared.

Patient and Public Involvement statement This review will not recruit patients, and they will not directly be involved in the design and implementation of this study.

Patient consent for publication Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

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Data sharing No additional data available.

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Figure 1. Flow chart of the study.

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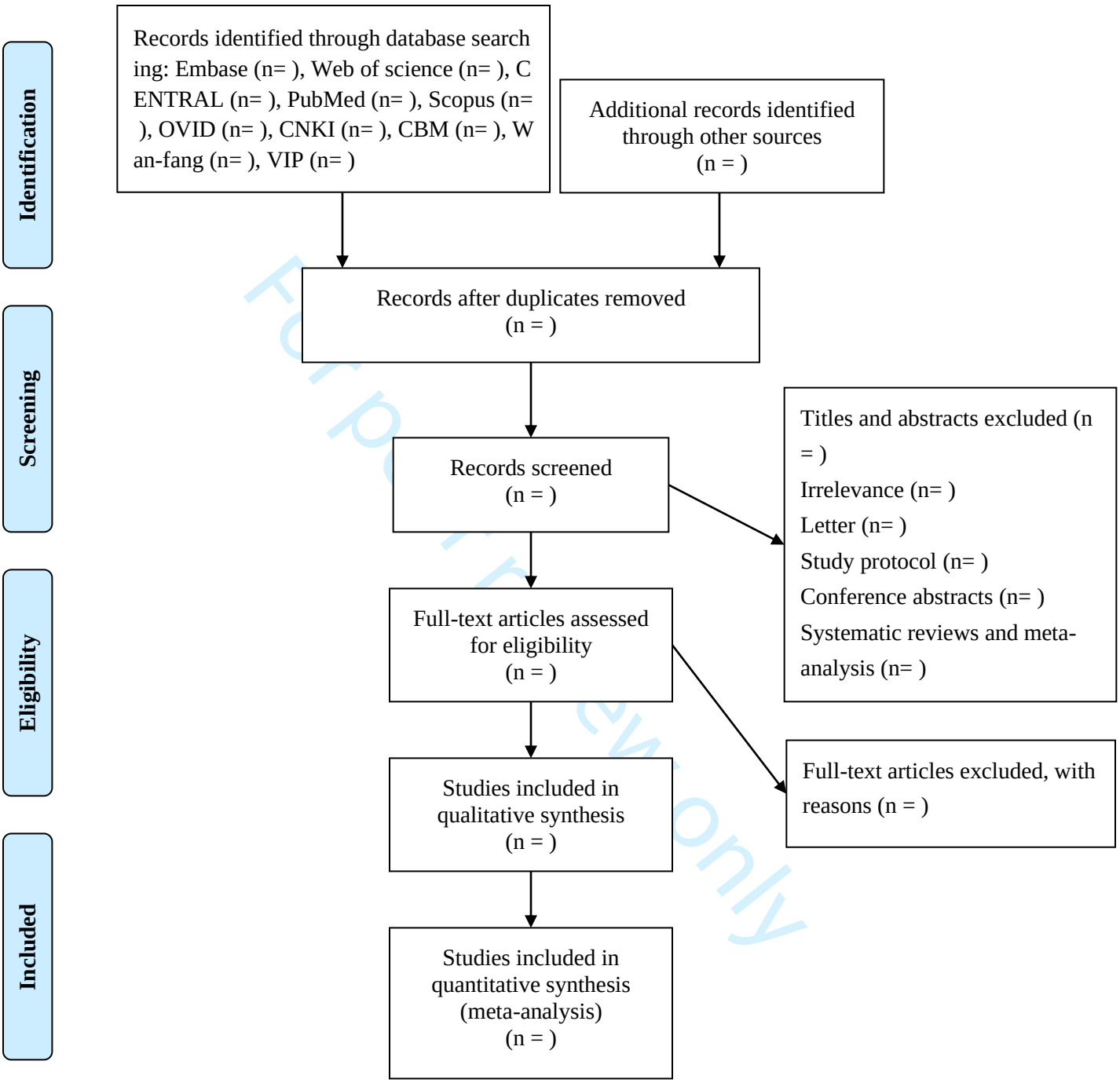


Figure 1. Flow chart of the study

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	1,8
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4,8
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1,2
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	19
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	19
Sponsor	5b	Provide name for the review funder and/or sponsor	19
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	19
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	6-8
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	8
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	8-10
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	10-11
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	11-12

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	12-13
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	12
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	12-13
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	13
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	13-14
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	14-16
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	14-16
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	17
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	16
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	17
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17-18

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Tai Chi and other mind-body interventions for cancer-related fatigue: an updated systematic review and network meta-analyses protocol

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ABSTRACT

Introduction

Fatigue is one of the most common symptoms in patients with cancer and is responsible for a reduced quality of life. There is a strong evidence base for mind-body interventions (MBIs) to manage cancer-related fatigue (CRF). However, the efficacy of Tai Chi and other MBIs in the treatment of CRF remains controversial.

Methods and analysis

We will perform a systematic review and network meta-analyses (NMA) that aim to assess the effects of Tai Chi and other MBIs in patients with CRF. The following databases will be searched from their inception to 1 August 2021: PubMed, EMBASE, Scopus, OVID, Web of Science, Cochrane Central Register of Controlled Trials (CENTRAL), the China National Knowledge Infrastructure, China Science and Technology Journal Database, Chinese Biomedical Database, and Wan Fang Digital Journals. We will include randomised controlled trials (RCTs) that compare MBIs with no treatment, placebo, and usual care in the treatment of CRF. The primary outcome will be changes in the fatigue state as evaluated by validated scales. We will perform a Bayesian NMA to analyse all the evidence for each outcome. The surface under the cumulative ranking curve and the mean ranks will be used to rank the various treatments. We will assess the quality of evidence contributing to network estimates of

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outcomes using the Grading of Recommendations Assessment, Development and Evaluation System framework.

Ethics and dissemination

This network meta-analyses will be disseminated through publication in a peer-reviewed journal. Since no individual patient data will be involved in the review, ethics approval and concerns about privacy are not be needed.

Trial registration number：CRD42021244999.

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Strengths and limitations of this study

- This study will include the latest randomised controlled trials to update the evidence base and obtain a comprehensive ranking of all included treatments.
- The revised Cochrane risk of bias tool will be used to assess the risk of bias in eligible studies.
- Grading of Recommendations Assessment, Development, and Evaluation system will be used to assess the quality of evidence.
- Different types of mind-body interventions (MBIs) may cause considerable heterogeneity in this review.
- Some MBIs, such as acupuncture and massage, will not be included in this review; this may affect the results.

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INTRODUCTION

Cancer-related fatigue (CRF) is defined as "a distressing, persistent, subjective sense of physical, emotional, and/or cognitive tiredness or exhaustion related to cancer or cancer treatment that is not proportional to recent activity and which interferes with usual functioning" by the National Comprehensive Cancer Network (NCCN).[1] CRF is one of the most prevalent and distressing symptoms of cancer, and it might persist for years after treatment completion in survivors.[2,3] CRF estimates range from 14.03% to 100% depending on the latest research.[4,5] Furthermore, CRF has a significant effect on physical functioning during treatment, and it is uncertain whether patients regain full functioning after completion of treatment. Persistent CRF causes disruption in all aspects of quality of life (QoL) during and after treatment.[6,7] Furthermore, CRF can cause difficulties in end-of-life care and it might be a risk factor for reduced survival.[1]

The pathophysiology of CRF remains unknown. The proposed underlying CRF mechanisms include mitochondrial dysfunction, peripheral immune activation, inflammation dysfunction, and central mechanisms (neuropeptide, neurotransmitter, hypothalamic-pituitary-adrenal axis dysfunction).[8,9] To our knowledge, there is no gold standard for the management of CRF. Based on the NCCN for CRF, some nonpharmacologic interventions have shown a strong evidence base for treating CRF.[1] Interestingly, there is increasing interest in mind-body interventions

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(MBIs), such as Tai Chi, in oncology settings, which may provide a new and effective treatment for CRF.[10]

The National Center for Complementary and Integrative Health defined MBIs as "techniques designed to enhance the mind's capacity to affect bodily function and symptoms."[11] MBIs have been shown to be effective in decreasing the expression of inflammation-related genes and in reducing common cancer-related side effects, especially in alleviating fatigue symptoms.[12-14] Tai Chi is rooted in traditional Chinese medicine and has been practiced for several millennia. This complex, multicomponent mind-body intervention merges physical, spiritual, psychosocial, and behavioural elements to promote human health.[15] Randomised controlled trials (RCTs) have indicated that Tai Chi significantly alleviates fatigue in patients with lung cancer and breast cancer over time.[16,17]

However, the effect of Tai Chi on CRF is still controversial based on evidence from several systematic reviews and meta-analyses.[18-20] The inconsistent results may be due to differences in search strategies, inclusion criteria, and comparators in these reviews. Furthermore, published meta-analyses only reviewed trials before 2016. Recently, new RCTs have been conducted and published.[21-23] To provide comprehensive evidence for the treatment of CRF, it is necessary to re-evaluate the effectiveness of Tai Chi and other MBIs for CRF based on the latest resources available.

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Objectives

We aimed to collect RCTs comparing Tai Chi and other MBIs with placebo or other non-MBIs among CRF patients, and to conduct network meta-analyses (NMAs) to assess the comparative effects of Tai Chi and other MBIs on CRF.

METHODS AND ANALYSIS

The study will be conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis for systematic review protocols (PRISMA-P)[24] (http://www.prisma-statement.org/Extensions/Protocols.aspx) and NMA checklist[25] (http://www.prisma-statement.org/Extensions/NetworkMetaAnalysis.aspx). This review protocol has been registered in PROSPERO with a registration number CRD42021244999.

Criteria for consideration of studies in this review

Types of studies

RCTs on Tai Chi and other MBIs for CRF with fatigue outcomes will be included in this review. No limits will be applied to the language, publication status, and publication date of these studies.

Types of participants

Patients aged 18 years or older, of both sexes, with a diagnosis of CRF will be considered. No restrictions will be set for cancer type, cancer grade, during treatment, or after end of treatment

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in patients.

Types of interventions

We focus here on four types of MBIs that have received considerable research attention and are widely available to clinical and community populations: Tai Chi (such as Yang-style TC, Chen-style TC, Wu-style TC, Sun-style TC, 24 simplified TC, or movements of TC), Qigong (including Baduanjin, Yijinjing, and Wuqinxi), yoga, and meditation. In this study, all different types of Tai Chi and Qi gong will first be reviewed as the same intervention and compared with other control measures. Then, different types of Tai Chi and Qi gong will be analysed again as different interventions.

Types of comparators

To determine whether the effects of Tai Chi and Qigong are primarily due to physical activity and whether the effects of yoga are primarily due to stretching, exercise and stretching will be included as a control group.

We included and classified the comparators in studies as follows:

1. Tai Chi and other MBIs versus exercise/stretching/placebo therapies.

2. Tai Chi and other MBIs versus waiting list/no treatment/usual care.

3. MBIs versus MBIs.

Types of Outcome measures

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Primary outcome

We will extract the effect sizes at the first time point after the end of the interventions and the subsequent follow-up time points. Only RCTs that include the following primary outcomes, namely assessments of CRF using effective and validated scales such as Brief Fatigue Inventory, Multidimensional Fatigue Symptom Inventory-Short Form, Functional Assessment of Chronic Illness Therapy-Fatigue Survey, and Functional Assessment of Cancer Therapy-Fatigue, will be included in this review.

Secondary outcomes

The secondary outcomes will include QoL and adverse events. Questionnaires will be used to assess QoL. Questionnaires will be used for assessment of QoL, such as the Functional Assessment of Cancer Therapy, European Organization for Research and Treatment of Cancer Quality of Life Questionnaire Core 30, Quality of Life Questionnaire Breast Cancer Module 23, 36-Item Short Form Health Survey, Quality of Life in Adult Cancer Survivors, and World Health Organization Quality of Life Questionnaire.

Search methods

Published RCTs will be searched in the following electronic databases: PubMed, EMBASE, Scopus, OVID, Web of Science, Cochrane Central Register of Controlled Trials (CENTRAL), the China National Knowledge Infrastructure, China Science and Technology Journal Database,

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Chinese Biomedical Databases, and Wan Fang Digital Journals from the date of their inception to 1 August 2021 with no restrictions on language. Unpublished trials will be retrieved from the following clinical trial registries: the NIH clinical registry ClinicalTrials. gov, Australian New Zealand Clinical Trials Registry, and Chinese Clinical Registry. In addition, the reference lists of all relevant articles will be checked to identify additional studies.

The medical search headings terms (MeSH) and their synonyms (free text) will be combined using the Boolean operators "AND" and "OR". The following MeSH terms will be used: "cancer", "tumor", "carcinoma", "neoplasm", "fatigue", "Tai Chi", "Qi gong", "Yoga", "meditation", and "randomized controlled trial". The search strategies are shown in **Supplementary File**.

Data collection and analysis

Selection of studies

Two researchers (HJF and HZ) will independently select studies using the bibliographic software EndNote (https://www.endnote.com/). Initially, duplicate studies will be filtered out. The researchers will then read the titles and abstracts to exclude studies that do not satisfy the eligibility criteria. After this selection, the full text of all remaining articles will be extracted to determine which articles can be included. In the case of duplicate studies, we will include only those trials with the most informative data. A third researcher (QWH) is responsible for

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reaching a consensus with the two authors on the inclusion or exclusion of each study. Finally, we will provide a list of excluded studies and justify the exclusions. The selection procedure will be shown in a PRISMA flow chart (**Figure 1**)

Data extraction

Two researchers (YT and DZ) will independently extract data using ADDIS (http://www.drugis.org/index) and Excel software with respect to five main domains: study information (e.g. title, source of publication, year of publication, first author's name and affiliation, country), participant information (e.g. gender, age, setting, cancer type, tumor grade, and basic cancer treatment plan), intervention details (e.g. intervention type, duration and frequency), methodology information (study design, random sequence generation, allocation concealment, blinding, and other concerns about bias), and outcome measures. The two researchers will cross-check and ascertain data accuracy. Disagreements will be resolved by a third researcher (SYD).

Unit of analysis issues

This is not an individual participant data review, and all analyses will be based on aggregated outcome data from the included RCTs.

Dealing with missing data

We will send a request for missing data to the original investigators of the trial or the contact

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person recorded in the trial registry. In case of no reply, we will use the last observation carried forward imputation method and impute the missing data with replacement values (https://training.cochrane.org/handbook/current/chapter-06). If means, SDs, and numbers of patients in each arm are not reported, we will transform the recorded SEs, t statistics, or *p* values to SDs, according to the Cochrane Handbook.[26,27] If possible, we will perform sensitivity analyses to assess how sensitive the results are to reasonable changes in the assumptions that are made. The potential impact of missing data on the findings of the review will be addressed in the Discussion section.

Quality assessment

Two independent researchers (YT and JL) will assess the risk of bias of the included RCTs in accordance with the revised Cochrane risk of bias tool.[28-30] Each domain will be rated as low risk, some concern, and high risk, based on the five distinct domains: the randomisation process, deviations from the intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. We will also make an overall risk of bias judgement according to the following criteria:[31] (1) low risk – all domains for this result; (2) some concerns – in at least one domain for this result but not at high risk of bias for any domain; (3) high risk – in at least one domain for this result, or the study is judged to have some concerns for multiple domains in a way that substantially lowers confidence in the results. If the researchers disagree,

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a third researcher (CW) will resolve the differences.

Data Synthesis and Analysis

The evidence-base and information flow in the network

We will systematically and comprehensively describe the characteristics of all eligible trials. A network diagram will be used to present the available evidence.[32] The size of the nodes will represent the sample size of the included trials. The thickness of the line will reflect the number of studies for each direct comparison, and the colour of each edge will imply the risk of bias. For NMAs including many competing interventions and multi-arm studies, we will instead use a table to show the network structure.[33,34] To understand how much each of the direct comparisons contributes to the final summary data, the contribution matrix will be used to display the percentage information of the direct evidence contributing to each relative effect estimated for a study.[35]

Pairwise meta-analyses

We will use the DerSimonian-Laird random effects model for pair-wise comparison. Standardised mean differences (SMD) for continuous outcomes or odds ratios (ORs) for dichotomous outcomes, both with a 95% confidence interval (CI), will be calculated as effect measures. The Cochran's Q χ^2 test and I^2-statistic will be measured for heterogeneity as a measure to reflect the underlying differences between the RCTs that directly compare the same

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pair of interventions. A *p* value of up to 0.10, and I^2 value of above 50% will indicate high heterogeneity.[29] We will perform sensitivity analysis of pairwise meta-analyses to validate the robustness of the results by omitting studies with unacceptable sources of heterogeneity.

Examination of assumptions in network meta-analyses (transitivity, inconsistency, and heterogeneity)

Studies that compare different interventions may differ in a broad range of characteristics, which are sometimes associated with the effect of an intervention and are referred to as effect modifiers. We will evaluate the transitivity assumption underlying NMAs by comparing the distribution of clinical and methodological variables which can act as effect modifiers across treatment comparisons.[26,27,36] Consistency in NMA means that the different sources of evidence (direct and indirect) are consistent with each other. We will employ the node splitting method and heatmap to investigate the inconsistency of the model by separating evidence on a particular comparison into direct and indirect evidence.[27, 36, 37] The loop-specific approach will be used to evaluate the presence of inconsistency locally in each closed loop.[27, 36] We will also calculate the I^2-statistic to evaluate consistency and heterogeneity in the entire network.[27,36] If there is inconsistency and considerable heterogeneity in the RCTs, network meta-analyses will not be performed and a narrative systematic review will be provided instead.

Network meta-analyses

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All analyses will be performed using the *gemtc*, *netmeta*, and *ggplot* package of R version 3.5.0 and the *network* package in Stata version 15.1. If the included trials meet the above three assumptions, we will perform a random-effects NMA within a Bayesian framework. We will fit our model using WinBUGS (version 1.4, three Markov chains, 50 000 iterations, an initial burn-in of 10 000, and a thinning of 10) and use uninformative prior distributions for the treatment effects. The binomial likelihood will be used for dichotomous outcomes and the normal likelihood for continuous outcomes.[26,27] ORs or SMD for all pairwise comparisons with 95% CI will be summarised in a league table. We will use the surface under the cumulative ranking curve (SUCRA) and the mean ranks to rank the various interventions for all outcomes.[38] A SUCRA equal to 1 means that the treatment is considered to be the best, while 0 means that it is certain to be the worst.

Publication bias, subgroup analyses, and meta-regression

We will perform a contribution plot to evaluate the contribution of each direct comparison to the assessment of each network meta-analytic summary effect. Additionally, we will use comparison-adjusted funnel plots to detect the potential publication bias in the results between imprecise and more precise trials.[39] To assess whether the treatment effects for the primary outcome are impacted by effect modifiers, subgroup analyses and network meta-regression will be conducted according to the following characteristics: (1) cancer type; (2) patient status

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(ongoing or post cancer treatment or no treatment); (3) randomisation and blinding; and (4) sample size (fewer than 25 patients per intervention arm). The sensitivity analysis of NMA will be narrowed into head-to-head studies or trials with a low risk of bias.

Summary of evidence

Two experienced researchers (HZ and DZ) will independently assess the certainty of evidence contributing to each network estimate of the primary outcome using the grading of recommendations assessment, development, and evaluation framework. Evidence quality will be rated as high, moderate, low, or very low on the basis of the study limitations, imprecision, inconsistency, indirectness, and publication bias.[40] Discrepancies will be resolved by a third researcher (CW).

Patient and Public Involvement

This review will not recruit patients, and they will not directly be involved in the design and implementation of this study.

Ethics and dissemination

This network meta-analyses will be disseminated through publication in a peer-reviewed journal. Since no individual patient data will be involved in the review, ethics approval and concerns about privacy are not be needed.

DISCUSSION

Although the effect of Tai Chi on CRF is still controversial, Tai Chi has the potential to be an

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effective complementary treatment option for CRF. In this review, we will perform the most comprehensive and up-to-date literature search, including ten electronic databases and clinical research registration websites. We will also extract the effect sizes at the end of the intervention period and post-intervention follow-up to evaluate the short- and long-term efficacy. Pairwise and network meta-analyses will also be performed. We are confident that this research can update the evidence of Tai Chi and other MBIs in the treatment of CRF and provide evidence-based medicine for patients, clinicians, and policy makers.

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Contributors

HJF and HZ contributed equally. HJF, YT and CW conceived the review protocol. HJF drafted the manuscript. QWH and CW revised the study design. HJF, HZ and QWH will perform the study search and study selection. YT, DZ, and SYD will carry out the data collection. YT, JL, and CW will assess the quality of included RCTs. HJF and YT will conduct the data analysis. QWH and CW will monitor each procedure of the review and are responsible for the quality control. All authors have read and approved the publication of the protocol.

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Competing interest statement None declared.

Patient consent for publication Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

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Data sharing No additional data available.

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Figure 1. Flow chart of the study.

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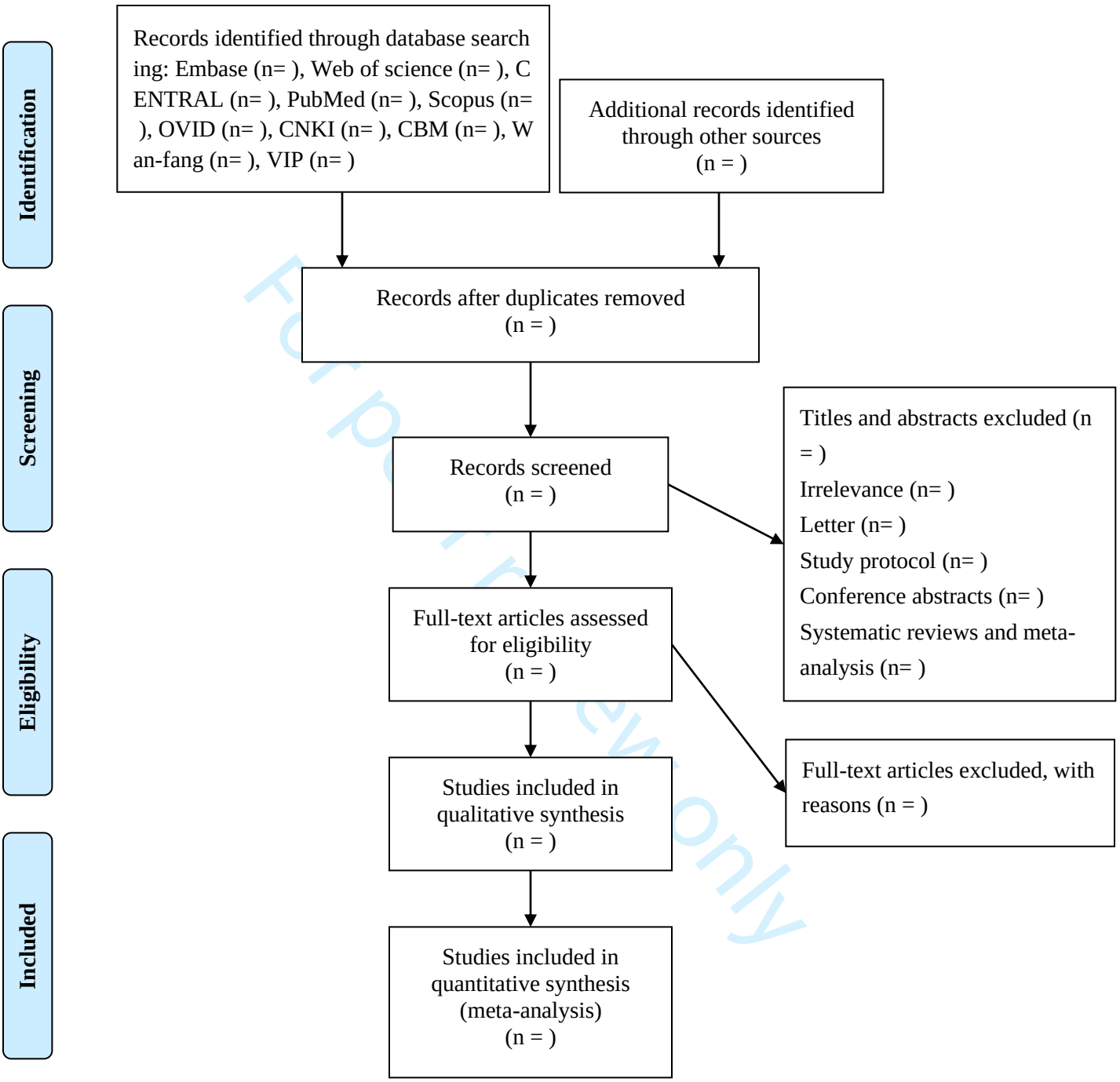


Figure 1. Flow chart of the study

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Search strategy

Database	#	Search strategy
1) Embase	#1	'neoplasm'/exp
	#2	cancer:ab,ti OR tumor:ab,ti OR neoplasm:ab,ti OR carcinoma:ab,ti
	#3	#1 OR #2
	#4	'fatigue'/exp
	#5	fatigue:ab,ti OR exhaustion:ab,ti OR lassitude:ab,ti OR tiredness:ab,ti OR weariness:ab,ti OR tired:ab,ti
	#6	#4 OR #5
	#7	#3 AND #6
	#8	'tai chi'/exp OR 'yoga'/exp OR 'qigong'/exp OR 'meditation'/exp
	#9	'tai chi':ab,ti OR 'tai ji':ab,ti OR taijiquan:ab,ti OR 'tai chi quan':ab,ti OR qigong:ab,ti OR yoga:ab,ti OR yijinjing:ab,ti OR baduanjin:ab,ti OR wuqinxi:ab,ti OR meditation:ab,ti OR mind-body:ab,ti OR 'mind and body':ab,ti
	#10	#8 OR #9
	#11	'randomized controlled trial'/exp OR 'controlled clinical trial'/exp
	#12	'randomized controlled trial':ab,ti OR 'controlled clinical trial':ab,ti
	#13	#11 OR #12
	#14	#7 AND #10 AND #13
2) Web of science	#1	**TOPIC:** (Cancer) *OR* **TOPIC:** (tumor) *OR* **TOPIC:** (neoplasm) *OR* **TOPIC:** (carcinoma) *Databases= WOS, CSCD, DIIDW, INSPEC, KJD, MEDLINE, RSCI, SCIELO Timespan=All years* *Search language=Auto*
	#2	**TOPIC:** (fatigue) *OR* **TOPIC:** (exhaustion) *OR* **TOPIC:** (lassitude) *OR* **TOPIC:** (tiredness) *OR* **TOPIC:** (weariness) *OR* **TOPIC:** (tired) *Databases= WOS, CSCD, DIIDW, INSPEC, KJD, MEDLINE, RSCI, SCIELO Timespan=All years* *Search language=Auto*
	#3	#1 AND #2
	#4	**TOPIC:** ("Tai Chi") *OR* **TOPIC:** ("Tai ji") *OR* **TOPIC:** ("Tai Chi Quan") *OR* **TOPIC:** (taijiquan) *OR* **TOPIC:** (qigong) *OR* **TOPIC:** (baduanjin) *OR* **TOPIC:** (yijinjing) *OR* **TOPIC:** (wuqinxi) *OR* **TOPIC:** (yoga) *OR* **TOPIC:** (meditation) *OR* **TOPIC:** (mind-body) *OR* **TOPIC:** ("mind and body") *Databases= WOS, CSCD, DIIDW, INSPEC, KJD, MEDLINE, RSCI, SCIELO Timespan=All years* *Search language=Auto*
	#5	**TOPIC:** ("Randomized controlled trial") *OR* **TOPIC:** ("Controlled trial") *OR* **TOPIC:** (randomized)

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		Databases= WOS, CSCD, DIIDW, INSPEC, KJD, MEDLINE, RSCI, SCI ELO Timespan=All years Search language=Auto
	#6	#3 AND #4 AND #5 *Databases= WOS, CSCD, DIIDW, INSPEC, KJD, MEDLINE, RSCI, SCI ELO Timespan=All years Search language=Auto*
3) Cochrane	#1	MeSH descriptor: [neoplasm] explode all trees
	#2	(cancer):ti,ab,kw OR (tumor):ti,ab,kw OR (neoplasm):ti,ab,kw OR (carcinoma):ti,ab,kw
	#3	#1 OR #2
	#4	MeSH descriptor: [fatigue] explode all trees
	#5	(fatigue):ti,ab,kw OR (exhaustion):ti,ab,kw OR (lassitude):ti,ab,kw OR (tiredness):ti,ab,kw OR (weariness):ti,ab,kw OR (tired):ti,ab,kw
	#6	#4 OR #5
	#7	#3 AND #6
	#8	MeSH descriptor: ["Tai ji"] explode all trees
	#9	MeSH descriptor: [Qigong] explode all trees
	#10	MeSH descriptor: [Yoga] explode all trees
	#11	MeSH descriptor: [Meditation] explode all trees
	#12	("Tai Chi"):ti,ab,kw OR ("Tai ji"):ti,ab,kw OR ("Tai Chi Quan"):ti,ab,kw OR (Taijiquan):ti,ab,kw OR (Qigong):ti,ab,kw OR (baduanjin):ti,ab,kw OR (yijinjing):ti,ab,kw OR (wuqinxi):ti,ab,kw OR (Yoga):ti,ab,kw OR (Meditation):ti,ab,kw
	#13	#8 OR #9 OR #10 OR #11 OR #12
	#14	MeSH descriptor: [Randomized Controlled Trials as Topic] explode all trees
	#15	("Randomized Controlled Trial"):ti,ab,kw OR ("Randomized"):ti,ab,kw OR ("Controlled"):ti,ab,kw
	#16	#14 OR #15
	#17	#7 AND #13 AND #16
4) PubMed	1	"Neoplasms"[MeSH Terms]
	2	"cancer"[Title/Abstract] OR "tumor"[Title/Abstract] OR "neoplasm"[Title/Abstract] OR "carcinoma"[Title/Abstract]
	3	#1 OR #2
	4	"Fatigue"[MeSH Terms]

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	5	"fatigue"[Title/Abstract] OR "exhaustion"[Title/Abstract] OR "lassitude"[Title/Abstract] OR "tiredness"[Title/Abstract] OR "weariness"[Title/Abstract] OR "tired"[Title/Abstract]
	6	#4 OR #5
	7	#3 AND #6
	8	"Tai ji"[MeSH Terms] OR "Qigong"[Publication Type] OR "Yoga"[Publication Type] OR "Meditation"[Publication Type]
	9	"Tai Chi"[Title/Abstract] OR "Tai ji"[Title/Abstract] OR "Tai Chi Quan"[Title/Abstract] OR "taijiquan"[Title/Abstract] OR "Qigong"[Title/Abstract] OR "baduanjin"[Title/Abstract] OR "yijinjing"[Title/Abstract] OR "wuqinxi"[Title/Abstract] OR "Yoga"[Title/Abstract] OR "Meditation"[Title/Abstract]
	10	#8 OR #9
	11	"Randomized Controlled Trials as Topic"[MeSH Terms] OR "Randomized Controlled Trial"[Publication Type]
	12	"Randomized Controlled Trials"[Title/Abstract] OR "Controlled Trials"[Title/Abstract] OR "ramdomized"[Title/Abstract] OR "clinical trial"[Title/Abstract] OR "clinical study"[Title/Abstract]
	13	#11 OR #12
	14	#7 AND #10 AND #13
5）Ovid	1	(cancer or tumor or carcinoma or neoplasm).ab.
	2	(fatigue or exhaustion or lassitude or tiredness or weariness or tired).ab.
	3	1 AND 2
	4	("tai chi" or "tai ji" or "tai chi quan" or "taijiquan" or Qigong or Baduanjin or Yijinjing or Wuqinxi or Yoga or Meditation).ab.
	5	("randomized controlled trial" or "controlled trial" or randomized).ab.
	6	3 AND 4 AND 5
6）Scopus	1	((TITLE-ABS-KEY (cancer) OR TITLE-ABS-KEY (tumor) OR TITLE-ABS-KEY (carcinoma) OR TITLE-ABS-KEY (neoplasm)) AND (TITLE-ABS-KEY (fatigue) OR TITLE-ABS-KEY (exhaustion) OR TITLE-ABS-KEY (lassitude) OR TITLE-ABS-KEY (tiredness) OR TITLE-ABS-KEY (weariness) OR TITLE-ABS-KEY (tired))) AND (TITLE-ABS-KEY ("tai chi") OR TITLE-ABS-KEY ("tai ji") OR TITLE-ABS-KEY "tai chi quan") OR TITLE-ABS-KEY (taijiquan) OR TITLE-ABS-KEY (qigong) OR TITLE-ABS-KEY (baduanjin) OR TITLE-ABS-KEY (yijinjing) OR TITLE-ABS-KEY (wuqinxi) OR TITLE-ABS-KEY (yoga) OR TITLE-ABS-KEY (meditation)) AND (TITLE-ABS-KEY ("randomized controlled trial") OR TITLE-ABS-KEY ("controlled trial") OR TITLE-ABS-KEY (rand

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		omized))
7） **CBM**	1	"Cancer" [common fields: intelligent] OR "carcinoma" [common fields: intelligent] OR "tumor" [common fields: intelligent] OR "neoplasm" [common fields: intelligent]
	2	"Tumor" [Unweighted: Extended]
	3	(#1) OR (#2)
	4	"Fatigue" [Unweighted: Extended]
	5	"Fatigue" [common fields: intelligent] OR "exhaustion" [common fields: intelligent] OR "lassitude" [common fields: intelligent] OR "tiredness" [common fields: intelligent] OR "weariness" [common fields: intelligent] OR "tired" [common fields: intelligent]
	6	(#4) OR (#5)
	7	(#3) AND (#6)
	8	"Tai ji" [Unweighted: Extended] OR "Qigong" [Unweighted: Extended] OR "Yoga" [Unweighted: Extended] OR "Meditation" [Unweighted: Extended]
	9	"Tai Chi" [common fields: intelligent] OR "Tai ji" [common fields: intelligent] OR "Taijiquan" [common fields: intelligent] OR "Qigong" [common fields: intelligent] OR "baduanjin" [common fields: intelligent] OR "yijinjing" [common fields: intelligent] OR "wuqinxi" [common fields: intelligent] OR "Yoga" [common fields: intelligent] OR "meditation" [common fields: intelligent]
	10	(#8) OR (#9)
	11	"clinical trial" [Unweighted: Extended] OR "randomized controlled trial" [Unweighted: Extended]
	12	"randomized controlled trial" [common fields: intelligent] OR "clinical trial" [common fields: intelligent] OR "clinical observation" [common fields: intelligent]
	13	(#11) OR (#12)
	14	(#7) AND (#10) AND (#3)
8) **CNKI**	1	((SU %= 'carcinoma' OR SU %= 'cancer' OR SU %= 'tumor' OR SU %= 'neoplasm') AND (SU %= fatigue' OR SU %= 'exhaustion' OR SU %= 'lassitude' OR SU %= 'tiredness' OR SU %= 'weariness' OR SU %= 'tired')) AND (SU %= 'tai chi' OR SU %= 'taijiquan' OR SU %= 'qigong' OR SU %= 'baduanjin' OR SU %= 'yijinjing' OR SU %= 'wuqinxi' OR SU %= 'yoga' OR SU %= 'meditation') ((SU %= '癌症' OR SU %= '肿瘤' OR SU %= '癌') AND (SU %= '疲劳' OR SU %= '疲乏' OR SU %= '疲倦')) AND ((SU %= '太极' OR SU %= '太极拳' OR SU %= '气功' OR SU %= '易筋经' OR SU %= '八

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		段锦' OR SU %= '五禽戏' OR SU %= '瑜伽' OR SU %= '冥想'))
9) WangFang	1	topic: ((("carcinoma" OR "cancer" OR "neoplasm" OR "tumor") AND topic: ("fatigue" OR "exhaustion" OR "lassitude" OR "tiredness" OR "weariness" OR "tired")) AND topic: ("Tai chi" OR "tai ji" OR "tai chi quan" OR "taijiquan" OR "Qigong" OR "baduanjin" OR "yijinjing" OR "wuqinxi" OR "Yoga" OR "Meditation") 主题: (("癌症" or "肿瘤" or "癌") and ("疲劳" or "疲乏" or "疲倦")) and ("太极" or "太极拳" or "气功" or "八段锦" or "易筋经" or "五禽戏" or "瑜伽" or "冥想")
10) VIP	1	(((M= “cancer” OR M= “carcinoma” OR M= “tumor” OR M= “neoplasm”) AND (M= “fatigue” OR M= “exhaustion” OR M= “lassitude” OR M= “tiredness” OR M= “weariness” OR M= “tired”)) AND (M= “tai chi” OR M= “tai ji” OR M= “taijiquan” OR M= “tai chi quan” OR M= “qigong” OR M= “baduanjin” OR M= “yijinjing” OR M= “wuqinxi” OR M= “yoga” OR M= “meditation”) AND (M= “randomized controlled trial” OR M= “clinical trial” OR M= “clinical observation”) ((M=(癌 OR 癌症 OR 肿瘤) AND M=疲乏 OR 疲劳 OR 疲倦)) AND (M=(太极 OR 太极拳 OR 气功 OR 八段锦 OR 易筋经 OR 五禽戏 OR 瑜伽 OR 冥想))
11) the NIH clinical registry ClinicalTrials.gov **https://www.clinicaltrials.gov/** Condition or disease: cancer-related fatigue Study Type: Interventional Studies (clinical trials) Study Results: All results Intervention/treatment: “Tai Chi” or “Tai ji” or “taijiquan” or “tai chi quan” or “qigong” or “baduanjin” or “yijinjing” or “wuqinxi” or “yoga” or “meditation”		
12）the Australian New Zealand Clinical Trials Registry **http://www.anzctr.org.au/** Registry: All Health condition(s) or problem(s) studied: cancer-related fatigue Description of intervention(s)/exposure: “Tai Chi” or “Tai ji” or “taijiquan” or “tai chi quan” or “qigong” or “baduanjin” or “yijinjing” or “wuqinxi” or “yoga” or “meditation” Study type: Interventional Allocation to intervention: randomised		
13） the Chinese clinical registry **http://www.chictr.org.cn/index.aspx** Target disease: cancer-related fatigue (癌因性疲劳 OR 癌因性疲乏) Study phase: All (全部) Study Type: Interventional Studies (干预性试验) Intervention: “Tai Chi” or “Tai ji” or “taijiquan” or “tai chi quan” or “qigong” or “baduanjin” or “yijinjing” or “wuqinxi” or “yoga” or “meditation” (太极 OR 太极拳 OR 气功 OR 八段锦 OR 易筋经 OR 五禽戏 OR 瑜伽 OR 冥想)		

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Reported on Page #
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	1
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4,8
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1,2
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	18
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	18-19
Sponsor	5b	Provide name for the review funder and/or sponsor	18-19
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	19
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	6-7
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	8
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	8-10
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	10-11
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	11

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	11-12
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	11-12
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	12
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	12
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	13-14
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	14-16
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	14-16
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	16-17
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	15
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	16-17
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	17

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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